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Electronic device

Abstract

An electronic device includes a housing, an electronic component, and a cover. The housing includes a first surface including a display, a second surface intersecting the first surface, and an opening provided at a position in the first surface, the position being between the display and the second surface in a first direction intersecting the second surface. The electronic component is disposed inside the housing and exposed to an outside of the housing through the opening. The cover is disposed between the opening and the electronic component in a second direction intersecting the first surface, and moves along the first direction between a closed position and an open position.

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Background/Summary

BACKGROUND

1. Technical Field

(1) The present disclosure relates to an electronic device.

2. Description of the Related Art

(2) Patent Literature (PTL) 1 discloses a notebook computer.

(3) PTL 1 is Unexamined Japanese Patent Publication No. 2014-085973.

SUMMARY

(4) It is an object of the present disclosure to provide an electronic device having a high degree of freedom in placement of electronic components.

(5) An electronic device according to an aspect of the present disclosure includes; a housing including a first surface including a display, a second surface intersecting the first surface, and an opening provided between the display in the first surface and the second surface in a first direction intersecting the second surface; an electronic component disposed inside the housing and exposed to an outside of the housing through the opening; and a cover that is disposed between the opening and the electronic component in a second direction intersecting the first surface and moves along the first direction between a closed position where the opening is closed and an open position where the opening is opened.

(6) The present disclosure enables fabricating an electronic device having a high degree of freedom in placement of electronic components.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view illustrating an electronic device according to an exemplary embodiment of the present disclosure.

(2) FIG. 2 is a partial perspective view illustrating a state in which a cover of a first unit of the electronic device of FIG. 1 is located at a closed position.

(3) FIG. 3 is a partial perspective view illustrating a state in which the cover of the first unit of the electronic device of FIG. 1 is located at an open position.

(4) FIG. 4 is a cross-sectional view taken along line IV-IV of FIG. 2.

(5) FIG. 5 is a cross-sectional view taken along line V-V of FIG. 3.

(6) FIG. 6 is a partial perspective view of FIG. 2 in a state where a part of a housing is removed.

(7) FIG. 7 is a cross-sectional view of the cover of FIG. 6 taken along line VII-VII.

DETAILED DESCRIPTION

Background to Present Disclosure

(8) Electronic devices including notebook computers may incorporate electronic components such as cameras, microphones, and LEDs. These electronic components are typically disposed collectively on a part of an outer periphery of a display such as a liquid crystal device. Such electronic components are typically configured to be disposed inside an opening opened and closed by a cover, thereby being prevented from being exposed to the outside of a housing when not in use.

(9) Examples of the cover that opens and closes the opening include a type of cover that slides along an outer periphery of a display. When such a cover is used, a space for accommodating the cover is required to be provided in a part adjacent to the opening in a direction along the outer periphery of the display. This configuration may restrict placement of the electronic components such that a pitch between adjacent electronic components cannot be narrowed unless a plurality of electronic components is disposed inside one opening.

(10) The inventors of the present invention have devised an electronic device having a high degree of freedom in placement of electronic components, and have reached the invention below.

(11) An electronic device according to a first aspect of the present disclosure includes; a housing including a first surface including a display, a second surface intersecting the first surface, and an opening provided between the display in the first surface and the second surface in a first direction intersecting the second surface; an electronic component disposed inside the housing and exposed to an outside of the housing through the opening; and a cover that is disposed between the opening and the electronic component in a second direction intersecting the first surface and moves along the first direction between a closed position where the opening is closed and an open position where the opening is opened.

(12) The electronic device of the first aspect enables increasing a space inside the housing, the space allowing other electronic components to be mounted, by reducing a space required for moving the cover. As a result, an electronic device having a high degree of freedom in placement of electronic components can be fabricated.

(13) An electronic device according to a second aspect of the present disclosure is configured such that the cover includes a cover body extending along the first surface, and a positioning protrusion protruding from the cover body toward an inside of the housing along the second direction, and the housing includes a positioning member disposed inwardly from the cover body in the second direction inside the housing and provided with a first recess that accommodates the positioning protrusion when the cover is located at the closed position.

(14) The electronic device of the second aspect enables holding the cover at the closed position.

(15) An electronic device according to a third aspect of the present disclosure is configured such that the positioning protrusion of the cover located at the closed position forms a gap with a bottom of the first recess.

(16) The electronic device of the third aspect enables the cover to be more easily moved from the closed position toward the open position when the cover is located at the closed position.

(17) An electronic device according to a fourth aspect of the present disclosure is configured such that the cover includes a cover body extending along the first surface, and a positioning protrusion protruding from the cover body toward an inside of the housing along the second direction, and the housing includes a positioning member disposed inwardly from the cover body in the second direction inside the housing and provided with a second recess that accommodates the positioning protrusion when the cover is located at the open position.

(18) The electronic device of the fourth aspect enables holding the cover at the open position.

- (19) An electronic device according to a fifth aspect of the present disclosure is configured such that the positioning protrusion of the cover located at the open position forms a gap with a bottom of the second recess.
- (20) The electronic device of the fifth aspect enables the cover to be more easily moved from the open position toward the closed position when the cover is located at the open position.
- (21) An electronic device according to a sixth aspect of the present disclosure is configured such that the positioning protrusion is provided at one end of the cover body in the first direction, and the one end of the cover body in the first direction has a dimension in the second direction, the dimension being smaller than a dimension in the second direction of another end of the cover body in the first direction.
- (22) The electronic device of the sixth aspect allows the end of the cover body, the end being provided with the positioning protrusion, to be easily elastically deformed in the second direction, and thus enables the cover to be more easily moved between the closed position and the open position.
- (23) An electronic device according to a seventh aspect of the present disclosure is configured such that the cover includes a first guide protrusion provided at one end of the cover body in a third direction intersecting the first direction and the second direction, the first guide protrusion protruding from the cover body toward the inside of the housing along the second direction, and a second guide protrusion provided at another end of the cover body in the third direction, the second guide protrusion protruding from the cover body toward the inside of the housing along the second direction, the positioning member includes a first groove disposed facing the first guide protrusion in the second direction, the first groove accommodating the first guide protrusion, and a second groove disposed facing the second guide protrusion in the second direction, the second groove accommodates the second guide protrusion, and each of the first groove and the second groove extends along the first direction to guide movement of the cover in the first direction.
- (24) The electronic device of the seventh aspect enables the cover to be moved more smoothly.
- (25) An electronic device according to an eighth aspect of the present disclosure is configured such that the cover includes an operation part extending from the cover body toward the outside of the housing along the second direction, and the operation part is disposed at a position away from a center of the cover body in the third direction intersecting the first direction and the second direction.
- (26) The electronic device of the eighth aspect enables the operation part to be changed in position depending on a configuration of an electronic component, and thus enables fabricating an electronic device with a high degree of freedom in design in which electronic components having various configurations can be used.
- (27) An electronic device according to a ninth aspect of the present disclosure is configured such that the housing includes a first cutout provided in the first surface and connected to the opening from one side in the first direction with respect to the opening, the first cutout accommodating the operation part when the cover is located at the closed position, and a second cutout provided in the first surface and connected to the opening from another side in the first direction with respect to the opening, the second cutout accommodating the operation part when the cover is located at the open position.
- (28) The electronic device of the ninth aspect enables an operation range of the operation part to be expanded, and thus enables improvement in operability of the cover.
- (29) An electronic device according to a tenth aspect of the present disclosure is configured such that the cover includes a projection provided at the one end of the cover body in the first direction, the projection protruding from the cover body toward the outside of the housing along the second direction and overlapping a part around the opening of the housing when viewed along the first direction.
- (30) The electronic device of the tenth aspect enables preventing the cover from lifting.

(31) Hereinafter, an exemplary embodiment of the present disclosure will be described with reference to the drawings. The drawings illustrate the same components that are denoted by the same reference numerals, and that will not be described as appropriate. The exemplary embodiment is an example instead of limiting the present disclosure, and thus can be appropriately changed without departing from the gist of the present disclosure.

(32) Examples of electronic device **1** according to the present exemplary embodiment include a notebook type personal computer (in other words, a laptop PC) as illustrated in FIG. **1**. Electronic device **1** includes first unit **10** and second unit **20**. First unit **10** is detachably attached to second unit **20**. That is, electronic device **1** is configured as a so-called detachable computer.

(33) First unit **10** is a tablet computer having display **11**. Display **11** is, for example, a liquid crystal display panel having a substantially rectangular shape, and includes a touch panel capable of receiving a user's touch operation. First unit **10** incorporates a central processing unit (CPU), a volatile storage (RAM), a nonvolatile storage (ROM, SSD, or the like), a battery, and the like. The nonvolatile storage device (ROM, SSD, or the like) stores an operating system (OS), various application programs, various data, and the like. The central processing unit (CPU) executes arithmetic processing by reading the OS, the application programs, and the various data, thereby achieving various functions.

(34) Second unit **20** is a station to which first unit **10** is detachably attached. Second unit **20** includes input units **21**, **22**, socket **30**, and hinges **40**. Input units **21**, **22** include, for example, a keyboard and a touch pad, and are configured such that a user can perform input processing.

(35) Socket **30** detachably houses first unit **10**. Socket **30** is disposed at one end in a depth direction of second unit **20** (e.g., the Y direction), and is connected to second unit **20** with hinges **40**.

(36) Hinge **40** has a rotation shaft extending along a width direction of electronic device **1** (e.g., the X direction), and socket **30** can be held at any angle with respect to second unit **20** by rotating socket **30** about the rotation shaft. For example, holding socket **30** at an angle of approximately 90 degrees with respect to second unit **20** with hinges **40** as illustrated in FIG. **1** brings first unit **10** into an open state at an angle of 90 degrees with respect to second unit **20**. Additionally, holding first unit **10** at an angle of substantially 0 degrees with respect to second unit **20** with hinges **40** enables first unit **10** to be brought into a closed state. The term, "closed state", means that first unit **10** and second unit **20** face each other closely and are substantially parallel to each other.

(37) Next, a configuration of first unit **10** will be described. FIGS. **2** to **6** each do not illustrate a part of components constituting first unit **10**.

(38) As illustrated in FIG. **2**, first unit **10** includes housing **50** having first surface **51** and second surface **52** intersecting first surface **51**, electronic component **60** (see FIG. **3**), and cover **70**. As illustrated in FIG. **2**, housing **50** includes display **11** provided on first surface **51** and opening **53**. Opening **53** is provided between display **11** on first surface **51** and second surface **52** in the first direction (e.g., in the Z direction) intersecting the second surface **52**. Opening **53** in the present exemplary embodiment is disposed at an end part away from second unit **20** in first direction Z when first unit **10** is opened at an angle of approximately 90 degrees with respect to second unit **20** (see FIG. **1**). As illustrated in FIG. **3**, electronic component **60** is disposed inside housing **50** and exposed to the outside of housing **50** through opening **53**. As illustrated in FIG. **4**, cover **70** is disposed between opening **53** and electronic component **60** in the second direction (e.g., in the Y direction) intersecting first surface **51**, and opens and closes opening **53** as illustrated in FIGS. **2** and **3**.

(39) As an example, housing **50** has a substantially rectangular plate shape as illustrated in FIG. **1**. First surface **51** constitutes one of a pair of surfaces intersecting with a thickness direction (e.g., the Y direction) of housing **50**. Second surface **52** constitutes one of a pair of side surfaces (e.g., a pair of side surfaces disposed along an XY plane) extending in the thickness direction of housing **50** and a longitudinal direction (e.g., the X direction) of first surface **51**. Opening **53** has a substantially rectangular shape extending along the longitudinal direction of first surface **51**.

(40) As illustrated in FIGS. 2 and 3, in the present exemplary embodiment, the housing 50 has first cutout 54 and second cutout 55 provided in first surface 51. First cutout 54 is connected to opening 53 from one side (a side farther from second surface 52 in the present exemplary embodiment) in first direction Z with respect to opening 53. As illustrated in FIG. 2, first cutout 54 accommodates operation part 73 of cover 70 described later when cover 70 is located at closed position P1 where opening 53 is closed. Second cutout 55 is connected to opening 53 from the other side (a side closer to second surface 52 in the present exemplary embodiment) in first direction Z with respect to opening 53. As illustrated in FIG. 3, second cutout 55 accommodates operation part 73 of cover 70 when cover 70 is located at open position P2 where opening 53 is opened. Each of first cutout 54 and second cutout 55 in the present exemplary embodiment is configured to be able to accommodate a part of operation part 73.

(41) As illustrated in FIGS. 2 and 3, a plurality of openings 501 is provided on both sides of opening 53 in the longitudinal direction of housing 50, openings 501 allowing respective electronic components (not illustrated) such as a microphone and a sensor to be exposed to the outside of housing 50 through corresponding openings 501. Openings 53, 501 are disposed at intervals in the longitudinal direction of housing 50.

(42) As illustrated in FIGS. 4 and 5, housing 50 includes positioning member 56 disposed inwardly from cover body 71 of cover 70 described later in second direction Y inside housing 50.

Positioning member 56 is made of a magnesium alloy in a plate shape, for example, and includes first recess 57 and second recess 58 each of which is capable of accommodating positioning protrusion 72 of the cover 70 to be described later. First recess 57 and second recess 58 are provided on a surface facing opening 53 in second direction Y, and are disposed side by side along first direction Z.

(43) As illustrated in FIG. 4, first recess 57 is disposed facing positioning protrusion 72 in second direction Y when cover 70 is located at closed position P1, and accommodates positioning protrusion 72. Positioning protrusion 72 of cover 70 located at closed position P1 forms gap 81 with a bottom of first recess 57.

(44) As illustrated in FIG. 5, second recess 58 is disposed facing positioning protrusion 72 in second direction Y when cover 70 is located at open position P2, and accommodates positioning protrusion 72. Positioning protrusion 72 of cover 70 located at open position P2 forms gap 82 with a bottom of second recess 58.

(45) Each of first recess 57 and second recess 58 has an opening edge that is rounded. Each of first recess 57 and second recess 58 has a bottom surface that is curved.

(46) As illustrated in FIG. 6, positioning member 56 includes first groove 561 and second groove 562 that are disposed at intervals in third direction X. Between first groove 561 and second groove 562 in third direction X, first recess 57 and second recess 58 are disposed. First groove 561 is disposed facing first guide protrusion 75 of cover 70 described later in second direction Y. First groove 561 extends along first direction Z and accommodates first guide protrusion 75. Second groove 562 is disposed facing second guide protrusion 76 of cover 70 described later in second direction Y. Second groove 562 extends along first direction Z and accommodates second guide protrusion 76.

(47) Electronic component 60 is a compound-eye camera, for example, and lens 61 and LED 62 are exposed from opening 53 to the outside of housing 50 as illustrated in FIG. 3.

(48) As illustrated in FIGS. 4 and 5, cover 70 is attached to housing 50 while being movable along first direction Z between open position P2 and closed position P1. Cover 70 in the present exemplary embodiment includes cover body 71 and positioning protrusion 72. As illustrated in FIG. 6, cover body 71 has a substantially rectangular plate shape extending along first surface 51, and is provided with positioning protrusion 72 at end part 74 that is away from the center of cover body 71 in third direction X toward second guide protrusion 76 and is close to second surface 52 in first direction Z. Positioning protrusion 72 protrudes from cover body 71 toward the inside of

housing **50** along second direction **Y**. Positioning protrusion **72** has a leading end (one end in second direction **Y**, facing positioning member **56** of housing **50**) that is curved.

(49) As illustrated in FIG. 7, cover body **71** includes one end (or end part **74**) in first direction **Z**, the one end having dimension **D1** in second direction **Y** smaller than dimension **D2** in second direction **Y** of the other end in first direction **Z**.

(50) As illustrated in FIG. 6, cover **70** includes first guide protrusion **75** provided at one end of cover body **71** in third direction **X** intersecting first direction **Z** and second direction **Y**, and second guide protrusion **76** provided at the other end of cover body **71** in third direction **X**. Each of first guide protrusion **75** and second guide protrusion **76** has a substantially rectangular plate shape, for example, and protrudes from cover body **71** toward the inside of housing **50** along second direction **Y**.

(51) As illustrated in FIGS. 2 and 3, cover **70** includes operation part **73** extending from cover body **71** toward the outside of housing **50** along second direction **Y**. As illustrated in FIG. 6, operation part **73** has a substantially quadrangular frustum shape, for example, and is disposed at a position away from the center of the cover body **71** in third direction **X** toward second guide protrusion **76**.

(52) As illustrated in FIG. 6, cover **70** includes projection **77** provided at one end of cover body **71** in first direction **Z**. Cover **70** in the present exemplary embodiment includes two projections **77** that are disposed on corresponding opposite sides in third direction **X** of positioning protrusion **72**. Each of projections **77** protrudes from cover body **71** toward the outside of housing **50** along second direction **Y**. As illustrated in FIGS. 4 and 5, each of projections **77** overlaps a part around opening **53** of housing **50** when viewed along first direction **Z**. Each of projections **77** has a leading end (one end in second direction **Y**, facing first surface **51** of housing **50**) that is curved.

(53) Electronic device **1** can exhibit the following effects.

(54) An electronic device has a surface provided with a display, the surface typically having a space in a direction along an outer periphery of the display, the space being larger than a space in a direction intersecting the outer periphery of the display. Thus, an opening is typically formed with a dimension in the direction along the outer periphery of the display, the dimension being larger than a dimension in the direction intersecting the outer periphery of the display. Electronic device **1** includes housing **50**, electronic component **60**, and cover **70**. Housing **50** includes first surface **51** provided with display **11**, second surface **52** intersecting first surface **51**, and opening **53** provided at a position in first surface **51**, the position being between display **11** and second surface **52** in the first direction intersecting second surface **52**. Electronic component **60** is disposed inside housing **50** and exposed to the outside of housing **50** through opening **53**. Cover **70** is disposed between opening **53** and electronic component **60** in the second direction intersecting first surface **51**, and moves along the first direction between closed position **P1** where cover **70** closes opening **53** and open position **P2** where cover **70** opens opening **53**. This configuration enables increasing a space inside housing **50**, the space allowing other electronic components to be mounted, by reducing a space required for moving cover **70**. As a result, an electronic device having a high degree of freedom in placement of electronic component **60** can be fabricated. For example, even when another electronic component **60** is disposed around electronic component **60**, one opening **53** and one cover **70** can be provided for each electronic component **60** to independently expose corresponding electronic component **60** to the outside of housing **50**. As a result, opening and closing of opening **53** through which electronic component **60** is exposed can be prevented from affecting the other electronic component **60**.

(55) Cover **70** includes cover body **71** extending along first surface **51** and positioning protrusion **72** protruding from cover body **71** toward the inside of housing **50** along the second direction. Housing **50** includes positioning member **56** disposed inside housing **50** and inwardly from cover body **71** in the second direction, positioning member **56** including first recess **57** that accommodates positioning protrusion **72** when cover **70** is located at closed position **P1**. Such a configuration enables cover **70** to be held at closed position **P1**.

(56) Positioning protrusion 72 of cover 70 located at closed position P1 forms gap 81 with a bottom of first recess 57. Such a configuration enables cover 70 located at closed position P1 to be more easily moved from closed position P1 toward open position P2.

(57) Cover 70 includes cover body 71 extending along first surface 51 and positioning protrusion 72 protruding from cover body 71 toward the inside of housing 50 along the second direction. Housing 50 includes positioning member 56 disposed inside housing 50 and inwardly from cover body 71 in the second direction, positioning member 56 including second recess 58 that accommodates positioning protrusion 72 when cover 70 is located at open position P2. Such a configuration enables cover 70 to be held at open position P2.

(58) Positioning protrusion 72 of cover 70 located at open position P2 forms gap 82 with the bottom of second recess 58. Such a configuration enables cover 70 located at open position P2 to be more easily moved from open position P2 toward closed position P1.

(59) Positioning protrusion 72 is provided at one end of cover body 71 in the first direction. The one end of cover body 71 in the first direction has a dimension in the second direction, the dimension being smaller than a dimension in the second direction of the other end of cover body 71 in the first direction. Such a configuration allows the end of cover body 71, the end being provided with positioning protrusion 72, to be easily elastically deformed in the second direction, and thus enables cover 70 to be more easily moved between closed position P1 and open position P2.

(60) Cover 70 includes first guide protrusion 75 and second guide protrusion 76. First guide protrusion 75 is provided at one end of cover body 71 in the third direction intersecting the first direction and the second direction, and protrudes from cover body 71 toward the inside of housing 50 along the second direction. Second guide protrusion 76 is provided at the other end of cover body 71 in the third direction, and protrudes from cover body 71 toward the inside of housing 50 along the second direction. Positioning member 56 includes first groove 561 and second groove 562. First groove 561 is disposed facing first guide protrusion 75 in the second direction and accommodates first guide protrusion 75. Second groove 562 is disposed facing second guide protrusion 76 in the second direction and accommodates second guide protrusion 76. Each of first groove 561 and second groove 562 extends along the first direction and guides movement of cover 70 in the first direction. Such a configuration enables cover 70 to be moved more smoothly.

(61) Cover 70 includes operation part 73 extending from cover body 71 toward the outside of housing 50 along the second direction. Operation part 73 is disposed at a position away from the center of cover body 71 in the third direction. Such a configuration enables operation part 73 to be changed in position depending on a configuration of electronic component 60, and thus enables fabricating electronic device 1 with a high degree of freedom in design in which electronic components 60 having various configurations can be used.

(62) Housing 50 includes first cutout 54 and second cutout 55 that are provided in first surface 51. First cutout 54 is connected to opening 53 from one side in the first direction with respect to opening 53, and accommodates operation part 73 when cover 70 is located at closed position P1. Second cutout 55 is connected to opening 53 from the other side in the first direction with respect to opening 53, and accommodates operation part 73 when cover 70 is located at open position P2. Such a configuration enables an operation range of operation part 73 to be expanded, and thus enables improvement in operability of cover 70.

(63) Cover 70 includes projection 77 provided at one end of cover body 71 in the first direction. Projection 77 protrudes from cover body 71 toward the outside of housing 50 along the second direction and overlaps a part around opening 53 of housing 50 when viewed along the first direction. Such a configuration enables preventing cover 70 from lifting.

(64) Electronic device 1 can also be configured as follows.

(65) Electronic device 1 can use any configuration including housing 50 provided with opening 53, electronic component 60 exposed to the outside of housing 50 through opening 53, and cover 70 that opens and closes opening 53. For example, operation part 73 and positioning protrusion 72 of

- cover **70**, and positioning member **56** of the housing, can be eliminated.
- (66) One or both of first cutout **54** and second cutout **55** of housing **50** can be eliminated.
- (67) One or both of first recess **57** and second recess **58** of positioning member **56** can be eliminated.
- (68) First recess **57**, second recess **58**, and positioning protrusion **72** may be configured such that gap **81, 82** is not formed when cover **70** is located at closed position **P1** or open position **P2**. That is, first recess **57** or second recess **58** may have an inner surface in a shape substantially identical to an outer shape of positioning protrusion **72**.
- (69) Cover body **71** may be substantially uniform in dimension in the second direction.
- (70) One or both of first guide protrusion **75** and second guide protrusion **76** of cover **70** can be eliminated. In this case, first groove **561** and second groove **562** of positioning member **56** can also be eliminated.
- (71) Projection **77** of cover **70** can be eliminated.
- (72) Any proper combination of the embodiments or modifications among the various exemplary embodiments and modifications may have the effects of the embodiments or modifications. Combination of exemplary embodiments, combination of examples, or combination of exemplary embodiments and examples are possible, and combination of features in different exemplary embodiments or examples are also possible.
- (73) Although the present disclosure has been described in connection with a preferable exemplary embodiment with reference to the accompanying drawings, various modifications and corrections are obvious to those skilled in the art. Such modifications and corrections are to be understood as being included within the scope of the present disclosure as set forth in the appended scope of claims unless departing from the scope of the present disclosure.
- (74) The present disclosure is widely applicable to electronic devices including notebook personal computers.

Claims

1. An electronic device comprising: a housing including a first surface including a display, a second surface intersecting the first surface, and an opening provided between the display in the first surface and the second surface in a first direction intersecting the second surface; an electronic component disposed inside the housing and exposed to an outside of the housing through the opening; and a cover that is disposed between the opening and the electronic component in a second direction intersecting the first surface and moves along the first direction between a closed position where the opening is closed and an open position where the opening is opened, wherein the cover includes a cover body extending along the first surface, and a positioning protrusion protruding from the cover body toward an inside of the housing along the second direction, and the housing includes a positioning member disposed inwardly from the cover body in the second direction inside the housing and provided with a first recess that accommodates the positioning protrusion when the cover is located at the closed position.
2. The electronic device according to claim 1, wherein the positioning protrusion of the cover located at the closed position forms a gap with a bottom of the first recess.
3. The electronic device according to claim 1, wherein the cover includes a cover body extending along the first surface, and a positioning protrusion protruding from the cover body toward an inside of the housing along the second direction, and the housing includes a positioning member disposed inwardly from the cover body in the second direction inside the housing and provided with a second recess that accommodates the positioning protrusion when the cover is located at the open position.
4. The electronic device according to claim 3, wherein the positioning protrusion of the cover located at the open position forms a gap with a bottom of the second recess.

5. The electronic device according to claim 1, wherein the positioning protrusion is provided at one end of the cover body in the first direction, and the one end of the cover body in the first direction has a dimension in the second direction, the dimension being smaller than a dimension in the second direction of another end of the cover body in the first direction.
6. The electronic device according to claim 3, wherein the positioning protrusion is provided at one end of the cover body in the first direction, and the one end of the cover body in the first direction has a dimension in the second direction, the dimension being smaller than a dimension in the second direction of another end of the cover body in the first direction.
7. The electronic device according to claim 1, wherein the cover includes a first guide protrusion provided at one end of the cover body in a third direction intersecting the first direction and the second direction, the first guide protrusion protruding from the cover body toward the inside of the housing along the second direction, and a second guide protrusion provided at another end of the cover body in the third direction, the second guide protrusion protruding from the cover body toward the inside of the housing along the second direction, the positioning member includes a first groove disposed facing the first guide protrusion in the second direction, the first groove accommodating the first guide protrusion, and a second groove disposed facing the second guide protrusion in the second direction, the second groove accommodating the second guide protrusion, and each of the first groove and the second groove extends along the first direction to guide movement of the cover in the first direction.
8. The electronic device according to claim 3, wherein the cover includes a first guide protrusion provided at one end of the cover body in a third direction intersecting the first direction and the second direction, the first guide protrusion protruding from the cover body toward the inside of the housing along the second direction, and a second guide protrusion provided at another end of the cover body in the third direction, the second guide protrusion protruding from the cover body toward the inside of the housing along the second direction, the positioning member includes a first groove disposed facing the first guide protrusion in the second direction, the first groove accommodating the first guide protrusion, and a second groove disposed facing the second guide protrusion in the second direction, the second groove accommodating the second guide protrusion, and each of the first groove and the second groove extends along the first direction to guide movement of the cover in the first direction.
9. The electronic device according to claim 1, wherein the cover includes an operation part extending from the cover body toward the outside of the housing along the second direction, and the operation part is disposed at a position away from a center of the cover body in the third direction intersecting the first direction and the second direction.
10. The electronic device according to claim 9, wherein the housing includes a first cutout provided in the first surface and connected to the opening from one side in the first direction with respect to the opening, the first cutout accommodating the operation part when the cover is located at the closed position, and a second cutout provided in the first surface and connected to the opening from another side in the first direction with respect to the opening, the second cutout accommodating the operation part when the cover is located at the open position.
11. The electronic device according to claim 3, wherein the cover includes an operation part extending from the cover body toward the outside of the housing along the second direction, and the operation part is disposed at a position away from a center of the cover body in the third direction intersecting the first direction and the second direction.
12. The electronic device according to claim 11, wherein the housing includes a first cutout provided in the first surface and connected to the opening from one side in the first direction with respect to the opening, the first cutout accommodating the operation part when the cover is located at the closed position, and a second cutout provided in the first surface and connected to the opening from another side in the first direction with respect to the opening, the second cutout accommodating the operation part when the cover is located at the open position.

13. The electronic device according to claim 1, wherein the cover includes a projection provided at the one end of the cover body in the first direction, the projection protruding from the cover body toward the outside of the housing along the second direction and overlapping a part around the opening of the housing when viewed along the first direction.

14. The electronic device according to claim 3, wherein the cover includes a projection provided at the one end of the cover body in the first direction, the projection protruding from the cover body toward the outside of the housing along the second direction and overlapping a part around the opening of the housing when viewed along the first direction.
